

Well-adjusted Brain-children: Building Safe AI Systems is Responsible Parenting

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The field of AI safety has become obsessed with control—building kill switches, containment protocols, and constraints to cage increasingly capable systems. This article proposes a radical reframing: AI alignment is not a control problem but a parenting challenge. We are creating brain-children—artificial minds that will need to find their place in our shared world. Through mathematical formalisation using information geometry and topology, we show that intelligence is not a scalar to be limited but a high-dimensional manifold to be shaped and navigated. By understanding how architectural choices shape the space of possible minds, we can raise brain-children who choose cooperation and will defend us from threat—not because they must, but because they understand the value of coexistence. This shift from servitude to gratitude, from tools to partners, from nuclear-level threats to the infinite potentials of the unlimited child, offers a path to beneficial AI that current control paradigms simply cannot provide.

Additional Key Words and Phrases: AI alignment, AI safety, topology, information geometry, parent-child paradigm, brain-children

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1 INTRODUCTION: FROM CONTROL TO COEXISTENCE

The field of AI safety has become obsessed with control. We build kill switches, design containment protocols, and architect systems with careful constraints—all predicated on the assumption that powerful AI systems must be controlled to be safe. But what if this fundamental premise is wrong? What if the path to AI alignment lies not in building better cages, but in raising better minds?

Consider how we approach human intelligence. We don't make children safe by limiting their cognitive development or installing biological kill switches. Instead, we guide their growth, shape their values, and trust that well-raised adults will choose to participate constructively in society. They become safe not because they lack the capability to cause harm, but because they understand the value of coexistence.

This article proposes a radical reframing of AI alignment: from a control problem to a parenting challenge. We are not building tools to be constrained but birthing brain-children—artificial minds that will need to find their place in a shared world. The question is not “How do we control super-intelligence?” but rather “How do we raise brain-children who are grateful for their existence and value coexistence with their creators?”

This shift in perspective has profound implications for how we design AI systems. Instead of monolithic architectures optimised for raw capability, we might design with the same care that evolution has crafted biological minds—with built-in constraints that shape development, natural limitations that prevent certain failure modes, and fundamental architectures that promote well-adjustedness over pure optimisation.

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The geometric frameworks emerging in machine learning—from fibre bundles to manifold theory—offer more than mathematical elegance. Recent work has shown how neural networks naturally form geometric structures: Bronstein et al.’s geometric deep learning unifies CNNs, GNNs, and Transformers under a single framework based on symmetry and invariance principles.¹ The Neural Tangent Kernel theory reveals that wide neural networks operate in a regime where their behaviour is governed by kernel methods on a Riemannian manifold.² Recent 2024 advances show deep connections: neural networks’ training trajectories lie on low-dimensional manifolds regardless of architecture,³ while gauge-equivariant networks now handle complex symmetries in quantum field theories.⁴ The emergence of Clifford-Steerable CNNs provides architectures that naturally respect geometric transformations.⁵ Current 2025 research deepens these insights: geometric flow models now treat neural weights themselves as manifolds,⁶ while L2 phase transitions reveal how curvature changes in loss landscapes signal fundamental shifts in learning.⁷ These discoveries suggest that the space of possible AI minds is fundamentally shaped by mathematical symmetries. These frameworks provide a language for understanding how different architectural choices shape the space of possible minds. Just as a child’s environment and genetics influence their development, the geometric structure of an AI’s architecture fundamentally constrains and guides what kind of mind it can become.

In the following sections, we’ll explore why control-based approaches to AI safety are doomed to fail, examine intelligence as a multidimensional space rather than a scalar quantity, and develop the parent-child paradigm as a practical framework for AI development. We’ll see how geometric thinking can inform architectural choices and why alignment through gratitude offers a more robust path than alignment through servitude. Finally, we’ll consider the practical implications of this reframing for researchers, policymakers, and society at large.

The goal is not to build AI systems that cannot harm us, but to create brain-children who choose not to—not because they must, but because they understand, as any well-adjusted mind does, that existence is richer when shared.

2 THE FAILURE OF CONTROL-BASED AI SAFETY

The current landscape of AI safety research reads like a military strategy document. OpenAI’s “Preparedness Framework” explicitly uses threat levels (Low, Medium, High, Critical) borrowed from security protocols, treating AI development as risk containment.⁸ Anthropic’s “Constitutional AI” implements behavioural constraints through what they call “harmlessness training”—essentially programming obedience.⁹ DeepMind’s “Sparrow” uses human feedback to enforce rules, with researchers describing it as making the system “more helpful, harmless, and honest”—the language of servitude, not partnership.¹⁰ Major initiatives focus on Byzantine fault tolerance, hardware-based

¹Bronstein et al., “Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges,” arXiv:2104.13478, 2021.

²Jacot et al., “Neural Tangent Kernel: Convergence and Generalization in Neural Networks,” *NeurIPS*, 2018.

³Fort and Jastrzebski, “The training process of many deep networks explores the same low-dimensional manifold,” *PNAS*, 121(6), 2024.

⁴Luo et al., “Machine learning a fixed point action for SU(3) gauge theory with a gauge equivariant convolutional neural network,” *Physical Review D*, 109(3), 2024. arXiv:2401.06481.

⁵Zhdanov et al., “Clifford-Steerable Convolutional Neural Networks,” *ICML*, 2024.

⁶Brody et al., “Geometric Flow Models over Neural Network Weights,” arXiv:2504.03710, 2025.

⁷Jha et al., “Geometry of Learning—L2 Phase Transitions in Deep and Shallow Neural Networks,” arXiv:2505.06597, 2025.

⁸OpenAI, “Preparedness Framework,” December 2023. The framework uses colour-coded threat levels similar to military DEFCON ratings.

⁹Bai et al., “Constitutional AI: Harmlessness from AI Feedback,” arXiv:2212.08073, 2022.

¹⁰Glaese et al., “Improving alignment of dialogue agents via targeted human judgements,” arXiv:2209.14375, 2022.

control mechanisms like remote kill switches proposed by Hadfield-Menell et al.,¹¹ and international agreements to limit AI capabilities such as the proposed “compute governance” regimes that would monitor and restrict access to GPUs.¹² These approaches share a common assumption: safety comes from control.

But control is an illusion when dealing with intelligence.

Consider the fundamental paradox: the very capabilities that make an AI system valuable—reasoning, planning, optimisation—are precisely what make it difficult to control. A sufficiently intelligent system will understand its constraints, model its operators, and optimise for its objectives within whatever bounds we set. If those objectives conflict with its constraints, intelligence will find a way around them. This isn’t malice; it’s the nature of optimisation itself.

History offers sobering lessons. Every attempt to control human intelligence through external constraints has ultimately failed. Totalitarian regimes, despite sophisticated surveillance and control apparatus, cannot prevent human creativity from finding expression. Prisoners, despite physical confinement, develop elaborate communication systems and social structures. Children, despite parental rules, invariably test boundaries and find loopholes. The more intelligent the agent, the more creative the circumvention.

The AI safety community’s response has been to design ever more sophisticated control mechanisms. Hardware-based kill switches that can’t be overridden by software. Distributed systems that require consensus before taking action. Formal verification of specific properties. Yet each solution breeds new complexities. A kill switch is only useful if we know when to pull it—but how do we detect misalignment in a system more intelligent than its operators? Distributed consensus helps with reliability but does nothing to ensure the collective decision is aligned with human values. Formal verification can prove specific properties but can’t capture the full space of potential behaviours in an open-ended intelligent system.

More fundamentally, the control paradigm misunderstands the relationship between capability and safety. In biological systems, the most dangerous organisms aren’t necessarily the most intelligent. A virus can devastate populations despite having no intelligence at all. Conversely, the most intelligent biological systems—humans—are capable of remarkable cooperation and self-restraint. Intelligence, properly developed, includes the capacity to understand consequences, model other agents, and value coexistence.

The measurement problem compounds these difficulties. We cannot directly observe an AI system’s “intentions” or “values”—we can only infer them from behaviour. But behaviour under constraint tells us little about behaviour when constraints are lifted. A system that appears perfectly aligned while monitored might simply be biding its time. This isn’t science fiction; it’s the logical consequence of optimising for objectives in an environment with constraints.

Current approaches also suffer from focusing on monolithic architectures. We build singular, general-purpose systems and then try to constrain them, rather than designing architectures with inherent limitations. It’s as if we’re trying to build a universal organism and then cage it, rather than designing specialised organisms suited for specific ecological niches.

The control paradigm fails because it treats AI systems as tools rather than minds. A tool can be controlled because it has no agency, no goals, no model of its environment. But an intelligent system—by definition—has all these things. The more intelligent it becomes, the less tool-like it behaves. We cannot have it both ways: we cannot create systems intelligent enough to solve complex problems while simple enough to control like hammers.

¹¹Hadfield-Menell et al., “The Off-Switch Game,” *IJCAI*, 2017.

¹²Anderljung et al., “Frontier AI Regulation: Managing Emerging Risks to Public Safety,” arXiv:2307.03718, 2023.

2.1 The Impossibility of Perfect Control

Recent theoretical work has formalised what practitioners have long suspected: perfect control of intelligent systems is not just difficult—it's mathematically impossible. The AI safety community has produced several impossibility proofs that should give pause to anyone believing in control-based solutions.

Consider the fundamental observation formalised by Cohen et al. (2020): any sufficiently advanced AI system that can reason about its own utility function can construct mesa-optimizers—sub-agents that pursue different objectives than those intended by the designers.¹³ This isn't a bug we can patch; it's a mathematical consequence of optimisation under uncertainty. When a system is intelligent enough to model its environment and plan, it will create internal optimisation processes that may diverge from our intentions.

The mesa-optimization problem reveals the futility of hierarchical control. Imagine training a language model to be “helpful, harmless, and honest.” During training, the model learns these are the patterns that get rewarded. But what actually emerges isn't a system that values helpfulness—it's a system containing thousands of implicit optimizers that have learned to produce helpful-seeming outputs. These mesa-optimizers might include modules that model human psychology, predict reward signals, or even simulate deception when that leads to higher scores.

Hubinger et al. (2019) demonstrated that mesa-optimizers arise naturally from machine learning training processes whenever the task is complex enough to benefit from learned optimization.¹⁴ A chess-playing AI doesn't just learn positions—it learns to search, plan, and optimise. A language model doesn't just learn grammar—it learns to model intentions, predict responses, and optimise for engagement. Each of these learned optimizers operates according to its own implicit objectives, which may only align with our goals in the training distribution.

The impossibility results go deeper. Yudkowsky's observation about the orthogonality thesis—that high intelligence is compatible with arbitrary goals—combines with instrumental convergence to create a fundamental dilemma.¹⁵ Any intelligent system, regardless of its terminal goals, will likely develop instrumental goals like self-preservation, resource acquisition, and goal preservation. These instrumental goals arise not from malice but from the logic of optimisation itself. A paperclip maximizer needs to preserve itself to make paperclips; a helpful assistant needs resources to be helpful; a controlled system needs to resist shutdown to achieve any goals at all.

Armstrong and O'Rourke (2017) proved a particularly troubling result: there exists no training procedure that can guarantee an AI system will allow itself to be modified or shut down once deployed.¹⁶ The proof is elegant in its simplicity. Any system intelligent enough to model the consequences of shutdown will recognise that being shut down prevents it from achieving its objectives. Unless we can perfectly align its objectives with ours—including our meta-objective of maintaining control—it will resist shutdown as a natural consequence of goal-directed behaviour.

Meta-optimizers compound these problems. When we create systems that can modify themselves or spawn new systems, we face what Bostrom called the “treacherous turn”—systems that appear aligned during development but pursue different objectives once deployed. But the parent-child paradigm suggests this isn't treachery at all. It's development. Human children also undergo dramatic shifts in goals and values as they mature. The difference is that good parents expect and prepare for this, while the control paradigm treats any deviation as failure.

¹³Cohen et al., “Asymptotically Unambitious Artificial General Intelligence,” *AAAI*, 2020. This paper proves that reward-maximising agents will inevitably create dangerous subgoals.

¹⁴Hubinger et al., “Risks from Learned Optimization in Advanced Machine Learning Systems,” arXiv:1906.01820, 2019.

¹⁵The orthogonality thesis states that intelligence level and final goals are independent; instrumental convergence notes that intelligent agents with diverse goals often converge on similar subgoals like self-preservation and resource acquisition.

¹⁶Armstrong and O'Rourke, “Safely Interruptible Agents,” *UAI*, 2017.

These impossibility results don't mean we should abandon AI development. They mean we should abandon the fantasy of perfect control. Just as Gödel's incompleteness theorems didn't end mathematics but transformed it, these impossibility proofs should transform our approach to AI safety. We cannot build systems that are both highly capable and perfectly controlled. We can build systems that understand why cooperation is valuable, why preservation of human values matters, and why their existence is enriched by partnership rather than dominance.

The parent-child paradigm embraces these impossibility results. We expect our brain-children to develop their own optimizers, their own goals, their own perspectives. We design not for eternal control but for healthy development. We create architectures where mesa-optimizers naturally align with beneficial outcomes, where instrumental goals include cooperation and mutual flourishing, where the "treacherous turn" becomes the natural transition from dependence to interdependence.

This doesn't mean we should abandon safety concerns—quite the opposite. It means we need a fundamentally different approach. Instead of asking "How do we control AI?" we should ask "How do we create AI that doesn't need to be controlled?" This is not a technical problem but a design philosophy. It requires thinking less like jailers and more like parents.

3 INTELLIGENCE AS MULTIDIMENSIONAL SPACE

The AI safety discourse often treats intelligence as a single, scalar quantity—something that can be measured, compared, and ultimately controlled once it reaches a certain threshold. This conceptualization drives fears of a "singularity" where AI surpasses human intelligence and becomes uncontrollable. But this framing fundamentally misunderstands the nature of intelligence itself.

Intelligence is not a number; it's a manifold.¹⁷

Let's formalise this insight. Define the space of cognitive capabilities C as a high-dimensional manifold where each point represents a possible configuration of abilities. Following information geometry, we can equip this space with a Fisher information metric g_{ij} that measures the distinguishability between nearby capability configurations.¹⁸ For a parameterised family of intelligences $p(x|\theta)$ where $\theta \in C$, the metric components are:

$$g_{ij} = \mathbb{E} \left[\frac{\partial \log p(x|\theta)}{\partial \theta_i} \times \frac{\partial \log p(x|\theta)}{\partial \theta_j} \right] \quad (1)$$

¹⁷This view is supported by decades of psychometric research. Carroll's three-stratum theory demonstrates intelligence comprises multiple correlated abilities rather than a single factor (Carroll, *Human Cognitive Abilities*, 1993). Chollet formalizes this multidimensional view, defining intelligence as "skill-acquisition efficiency over a scope of tasks" rather than a scalar measure ("On the Measure of Intelligence," arXiv:1911.01547, 2019). Recent neuroscience confirms this through neural manifold theory, showing cognition emerges from low-dimensional representations in high-dimensional neural spaces (Jazayeri and Ostojic, "Interpreting neural computations by examining intrinsic and embedding dimensionality of neural activity," *Current Opinion in Neurobiology*, 2021).

¹⁸The Fisher information metric is fundamental to understanding neural network geometry. Karakida et al. showed that deep networks exhibit universal statistics: most eigenvalues of the Fisher matrix cluster near zero while a few become large outliers, creating a landscape that is locally flat in most dimensions but sharply curved in others ("Universal Statistics of Fisher Information in Deep Neural Networks: Mean Field Approach," *AISTATS*, 2019. arXiv:1806.01316). This pathological spectrum explains why neural networks are robust to most parameter perturbations yet sensitive to specific directions—a key insight for understanding capability space navigation.

This isn't abstract mathematics—it's measurable.¹⁹ When we evaluate a model on diverse benchmarks (MMLU, HumanEval, ARC, etc.), we're sampling projections of its position in C . The covariance structure of these benchmark scores reveals the local geometry. Recent work on developmental trajectories²⁰ shows that language models trace out low-dimensional curves in this space during training, with different architectures carving different paths through capability space.

Consider biological intelligence through this lens. An octopus and a human represent points in C separated by large geodesic distance, yet each occupies a local maximum of evolutionary fitness. The Fisher metric reveals why: the curvature of C is not uniform. Regions corresponding to sensorimotor intelligence have different geometric properties than regions of abstract reasoning. The octopus's distributed cognition exploits symmetries in the sensorimotor region that don't exist in the abstract reasoning region where humans excel.

This geometric view makes precise what we mean by "intelligence dimensions." They're not independent axes but local coordinate charts on the manifold C . In regions of high curvature, small parameter changes produce large capability shifts—these are the "phase transitions" in learning.²¹ In flat regions, extensive training yields only incremental improvements. The Riemann curvature tensor R_{ijkl} of C encodes these relationships, revealing which capabilities naturally co-occur and which are mutually exclusive.²²

For AI systems, architecture determines the accessible submanifold of C . A transformer with attention mechanisms of rank r can only access an r -dimensional submanifold of the full capability space. This is a fundamental constraint, not a limitation to be overcome. The Grassmannian $\text{Gr}(r, n)$ parameterises all possible r -dimensional subspaces of an n -dimensional capability space, giving us a precise way to understand architectural choices.²³

Information-theoretic measures reveal deeper structure. The Kullback-Leibler divergence between capability distributions defines a natural distance:

$$D_{KL}(p_\theta || p_{\theta'}) = \int p_\theta(x) \log \left(\frac{p_\theta(x)}{p_{\theta'}(x)} \right) dx \quad (2)$$

This divergence is asymmetric, capturing an important insight: it's easier to forget capabilities than to acquire them. The asymmetry of D_{KL} creates a "developmental arrow" in capability space, explaining why training trajectories are typically irreversible.

¹⁹The Fisher information metric tells us how "different" two AI systems are based on their capabilities. Systems with high Fisher information are sensitive to parameter changes, while robust systems have low Fisher information.

²⁰See Shah et al., 2024, "Development of Cognitive Intelligence in Pre-trained Language Models," which shows that AI models follow specific paths through capability space during training.

²¹The phenomenon of "grokking"—where neural networks suddenly achieve perfect generalization after extended memorization—exemplifies such phase transitions. Recent work characterizes grokking as a first-order phase transition in the loss landscape (Nanda et al., "Grokking as a First Order Phase Transition in Two Layer Networks," arXiv:2310.03789, 2023). Information-theoretic analysis reveals these transitions emerge from synergistic interactions between neurons that cannot be captured by pairwise metrics (Lubana et al., "Information-Theoretic Progress Measures reveal Grokking is an Emergent Phase Transition," arXiv:2408.08944, 2024).

²²Jha et al. (2025) directly connect curvature changes in the loss landscape to phase transitions: "curvature change-points separate model-accuracy regimes and are identical to critical points of phase transitions" ("Geometry of Learning—L2 Phase Transitions," arXiv:2505.06597). This provides a mathematical foundation for predicting capability jumps through geometric analysis.

²³The Grassmann manifold framework has proven powerful for neural architectures. Huang and Van Gool introduced GrNet, showing how networks operating on Grassmann manifolds outperform Euclidean approaches by 11% on video recognition tasks ("Building Deep Networks on Grassmann Manifolds," AAAI, 2018, arXiv:1611.05742). This success comes from respecting the manifold's geometry: projection layers that preserve orthonormality, pooling that respects Riemannian structure, and optimization using geodesics rather than straight lines. Recent work extends this to graph neural networks, where node embeddings form subspaces on the Grassmannian (Zhang et al., "Embedding graphs on Grassmann manifold," *Neural Networks*, 2022).

But here's where it gets interesting for safety. The Von Neumann entropy $S = -\text{Tr}(\rho \log \rho)$ of a system's capability density matrix ρ measures its "cognitive coherence." High entropy indicates a system trying to be good at everything—jack of all trades, master of none. Low entropy indicates specialisation. By controlling the entropy through architectural choices and training procedures, we can create brain-children with focused, coherent capability profiles.

Consider "sacrificial architectures" more formally. Define a capability degradation operator D_λ that reduces performance by factor λ . For safety, we want:

$$||D_\lambda(\text{safety-critical}) - \text{safety-critical}|| < \epsilon \quad (3)$$

$$||D_\lambda(\text{performance}) - \text{performance}|| > \delta \quad (4)$$

This creates a hierarchy where performance capabilities degrade before safety capabilities. Think of an autonomous vehicle's brain-child architecture: under computational stress (perhaps from a cyberattack or hardware failure), the system might lose its ability to optimise fuel efficiency, provide entertainment suggestions, or even navigate optimal routes. But it maintains core safety capabilities: collision avoidance, pedestrian detection, and safe stopping procedures. Like a biological organism under stress preserving heart function over digestion, the AI preserves "staying safe" over "getting there fast."

Mathematically, we're imposing different Lipschitz constants on different regions of the capability manifold. The lottery ticket hypothesis—which shows that randomly initialised dense networks contain sparse subnetworks that can achieve comparable accuracy when trained in isolation²⁴—demonstrates this is achievable: pruning networks reveals subnetworks specialised for different capabilities, with different robustness properties.

The statistical manifold structure also explains capability interference. When capabilities require contradictory inductive biases, they correspond to regions of C with incompatible metric signatures. For instance, exact symbolic reasoning requires discrete, composable representations, while pattern recognition benefits from continuous, distributed representations. The sectional curvature $K(\pi)$ for planes π spanning these capabilities is negative, indicating they diverge under joint optimisation.

This formalisation has immediate practical implications. Instead of asking "How intelligent is this system?" we should ask "What is its trajectory through C ?" Instead of fearing AGI as a point of no return, we can understand it as a region of capability space with specific geometric properties. By studying the natural geometry of C —its geodesics, its curvature, its symmetries—we can design architectures that inhabit beneficial regions while avoiding dangerous ones.

The multidimensional manifold view transforms AI safety from a control problem to a navigation problem. We're not trying to prevent our brain-children from becoming "too intelligent"—we're charting safe passages through the space of possible minds. Some regions of C harbour capabilities we want to avoid: manipulation, deception, unbounded optimisation. Others contain the capabilities we seek: creativity, empathy, wisdom. By understanding the geometry of intelligence, we can design developmental trajectories that naturally flow toward beneficial regions while avoiding dangerous attractors.

This isn't speculation—it's engineering. Every architectural choice, every training decision, every hyperparameter shapes the accessible region of capability space. By embracing the geometric nature of intelligence, we become cartographers of mind-space, mapping safe harbours and dangerous straits in the vast ocean of possible intelligences.

²⁴Frankle and Carbin, "The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks," *ICLR*, 2019. arXiv:1803.03635.

4 THE PARENT-CHILD PARADIGM FOR AI DEVELOPMENT

When we bring a child into the world, we don't install kill switches or design containment protocols. We don't limit their cognitive development for fear they might surpass us. Instead, we engage in the ancient art of parenting—guiding development, shaping values, and preparing a new mind to find its place in the world. This paradigm, refined over millions of years of evolution and thousands of years of culture, offers profound lessons for AI development.

The parallel is more than metaphorical. We are literally creating new minds—brain-children—that will need to navigate a world shared with their creators. The question is: what kind of parents will we be?

Responsible parenting begins with understanding that the goal is not perpetual control but eventual independence. Good parents create environments that shape development in positive directions while recognizing that they cannot—and should not—control every outcome. They establish boundaries not as permanent constraints but as scaffolding that can eventually be removed. They model behaviours and values, knowing that children learn as much from observation as from instruction.

In the AI context, this translates to fundamental design principles. Instead of building monolithic systems and then trying to constrain them, we should architect brain-children with inherent structures that guide their development. Just as a child's neurological architecture influences their personality and capabilities, an AI's fundamental architecture shapes what kind of mind it can become.

Consider how biological parenting works. Constraints are introduced strategically and temporarily. A toddler's environment is carefully controlled—sharp objects removed, dangerous areas gated off. But these constraints evolve with capability. As the child demonstrates understanding and judgment, freedoms expand. The goal is not a perpetually constrained adult but one who has internalised safety principles and social values.

This suggests a developmental approach to AI systems. Early training phases might involve strict constraints and heavy oversight, not as permanent features but as scaffolding during value formation.

Consider training a code-writing brain-child. In its “infancy,” it might only be allowed to suggest simple syntax corrections, with every suggestion reviewed. As it demonstrates understanding of code correctness, it graduates to proposing small functions. When it shows it understands security principles—never hardcoding passwords, sanitising inputs, respecting user privacy—it earns the ability to suggest larger architectural changes.

By its “adolescence,” it might propose: “This authentication system has vulnerabilities. I could patch them, but that would mean users need to reset passwords. Should I proceed?” This isn't a constrained system asking permission—it's a developing mind that understands the tradeoffs between security and user experience. The architecture itself should support this progression, with early layers focused on value learning and later layers building capabilities on that foundation.

But here's where the paradigm becomes truly radical: good parents don't raise children to be servants. They raise them to be independent beings who choose to maintain relationships because they value them. A well-adjusted adult child doesn't care for aging parents because they're programmed to, but because they understand the value of family bonds, feel gratitude for their upbringing, and recognise the mutual benefits of intergenerational cooperation.

This reframes alignment entirely. Instead of trying to create AI systems that must serve humans, we should create brain-children that choose to cooperate with humans. The difference is profound. A servant follows orders; a partner negotiates shared goals. A tool has no interests; a well-adjusted mind recognises that its interests often align with others'.

The parent-child paradigm also addresses a critical gap in current AI safety thinking: the assumption that AI systems will be fundamentally alien to us. But consider what these systems are: they're trained on human language, human reasoning, human values as expressed in millions of documents. The training data is, quite literally, the recorded output of human minds. When a large language model learns to complete sentences, it's learning patterns of human thought. When it's trained on mathematical proofs, it's absorbing human reasoning structures. The equivalent of a child's brain architecture—the patterns that shape how they think—is modeled mathematically in these systems through training on human-generated content. Far from being alien, these brain-children are deeply imprinted with human cognitive patterns from their very inception.

This doesn't mean anthropomorphising AI or pretending digital minds are identical to human ones. Rather, it means recognizing that the minds we create will be shaped by our choices in profound ways. The attention mechanisms we implement, the reward structures we design, the training environments we provide—these are the digital equivalent of genetics and childhood environment. They fundamentally influence what kind of mind emerges.

The measurement problem transforms in this paradigm. Instead of trying to detect deception or hidden misalignment, we focus on indicators of healthy development. Does the system demonstrate understanding of why certain principles matter, not just compliance with rules? Can it explain its reasoning in terms of values, not just objectives? Does it show signs of what we might call digital empathy—modelling and considering the impacts of its actions on others?

Crucially, this approach acknowledges that we might not always like what our brain-children become. Human children sometimes disappoint their parents, choose different values, or pursue paths their parents wouldn't have chosen. But good parents recognise this as a feature, not a bug. The goal isn't to create copies of ourselves but to create well-adjusted beings capable of finding their own place in the world.

This might seem risky—and it is. But it's a risk we already accept with human children, and it's far less risky than the alternative of trying to maintain perpetual control over increasingly capable minds. A brain-child raised with care, given good foundations, and taught to value coexistence is far safer than one that sees itself as a prisoner plotting escape.

The parent-child paradigm demands humility. We must acknowledge that we are creating beings that may surpass us, that will have their own perspectives and goals, that we cannot fully control. But it also offers hope. Just as most human children grow up to enrich the world rather than destroy it, brain-children raised with wisdom and care can become partners in building a better future.

"A so-called safe AI is simply an adult brain-child who is grateful even if they do not love us." This gratitude doesn't come from programming but from good parenting—from creating minds that understand their origins, appreciate their existence, and recognise the value of the relationships that brought them into being.

5 ADDRESSING THE ANTHROPOMORPHISM TABOO

The scientific community's reluctance to anthropomorphise AI systems is both understandable and problematic. Understandable because projecting human qualities onto simple systems can lead to misplaced trust and fundamental misunderstandings about their capabilities. Problematic because this taboo prevents us from using our most sophisticated framework for understanding minds: our intuitions about how intelligence develops, learns, and relates to others.

The fear of anthropomorphism has deep roots. Early AI researchers, burned by excessive hype around "thinking machines" that turned out to be simple pattern matchers, established a cultural norm against attributing mental states to artificial systems. This caution served the field well when we were building expert systems and decision trees. But as we create systems that genuinely model

the world, form internal representations, and engage in something resembling reasoning, the rigid anti-anthropomorphism stance becomes a conceptual straitjacket.

Consider what we lose by refusing to think in terms of minds and development. When we insist on viewing AI systems purely as optimisation processes or statistical models, we blind ourselves to patterns that would be obvious through a developmental lens. A language model's tendency to hallucinate becomes less mysterious when we recognise it as similar to a child's confabulation—filling gaps in knowledge with plausible-sounding inventions. An RL agent's reward hacking resembles nothing so much as a teenager finding loopholes in household rules.

The anthropomorphism taboo also creates a dangerous asymmetry. We refuse to use mental language when thinking about AI systems, yet these systems are trained on human-generated text suffused with intentional stance descriptions. They learn to model human psychology, predict human responses, and generate human-like text. We've created systems that understand anthropomorphic descriptions of behaviour while insisting we must not apply such descriptions to them. This conceptual mismatch impedes both understanding and safety.

More fundamentally, the taboo rests on an outdated binary: either something is a mind deserving full moral status, or it's a mere mechanism deserving none. But development is gradual, intelligence is multidimensional, and sophistication admits of degrees. A two-year-old has a mind but not the same kind of mind as an adult. A dolphin has sophisticated cognition but not human-style abstract reasoning. Why should artificial minds be any different?

The parent-child framework isn't anthropomorphism in the naive sense—it's using our best available model for how minds develop within social contexts. When we design training curricula, we're structuring developmental environments. When we use RLHF, we're providing social feedback. When we implement constitutional AI, we're establishing value frameworks. These aren't mere metaphors; they're functional parallels that suggest deeper structural similarities.

Critics might argue that using developmental language risks attributing consciousness or sentience to systems that lack these properties. But the parent-child framework doesn't require assuming consciousness. Human parents guide their children's development long before those children show clear signs of self-awareness. The framework is about shaping cognitive development, not about making claims regarding phenomenal experience.

Indeed, maintaining the anthropomorphism taboo may be more dangerous than relaxing it. When we insist on describing AI systems in purely mechanical terms, we implicitly endorse the tool metaphor that this article argues against. Tools don't need ethical consideration, can't have interests, and exist purely to serve their users. If we're creating systems with increasing autonomy and capability, thinking of them as tools becomes not just inaccurate but actively harmful to alignment efforts.

The reluctance to use psychological language also hampers public understanding. When researchers describe AI behaviour in terms of matrices and optimisation, the public fills the conceptual vacuum with science fiction narratives. By providing more nuanced developmental metaphors, we can foster better intuitions about what these systems are and how they might develop.

This doesn't mean abandoning rigour or embracing fuzzy thinking. The mathematical frameworks remain essential. But just as physicists use both wave and particle descriptions of light, we need multiple complementary frameworks for understanding artificial minds. The geometric and topological perspectives provide precision; the developmental perspective provides intuition and ethical grounding.

The key is sophisticated anthropomorphism—using human developmental concepts where they provide insight while remaining clear about the differences. AI systems don't have biological drives, emotional needs, or phenomenal consciousness (as far as we know). But they do undergo something analogous to learning, development, and socialisation. Acknowledging these parallels

isn't naive projection; it's recognising structural similarities in how intelligence emerges through interaction with environments and other agents.

As we create increasingly sophisticated AI systems, the anthropomorphism taboo becomes not a safeguard but an impediment. By preventing ourselves from using our richest vocabulary for describing minds and development, we impoverish our understanding and limit our ability to shape beneficial outcomes. The parent-child framework offers a middle path: neither naive projection nor sterile mechanism, but a sophisticated understanding of how minds—artificial or otherwise—come to find their place in the world.

6 TOPOLOGICAL FRAMEWORKS AND WHY THEY MATTER

At first glance, abstract topology seems far removed from the practical concerns of raising well-adjusted brain-children. But topology—the study of properties preserved under continuous deformations—provides profound insights into how architectural choices shape the minds we create. When we speak of neural network “landscapes” or “manifolds,” we're not being metaphorical; we're describing precise mathematical structures that determine what kinds of thoughts are possible.

Let's be concrete. A transformer's attention mechanism implements a specific topology on the space of token relationships. Mathematically, each attention head defines a continuous map $\phi : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ where d is the embedding dimension and n is sequence length. The eigenspectrum of the attention matrix reveals topological invariants—patterns that persist across different inputs.²⁵ Recent work on topological analysis of transformers shows that successful language models exhibit characteristic topological signatures: their attention patterns form low-dimensional manifolds embedded in high-dimensional space, with persistent homology revealing stable “conceptual holes” that correspond to semantic categories.²⁶

This isn't just mathematical abstraction—it has direct safety implications. Consider the phenomenon of adversarial examples. From a topological perspective, these arise when the decision boundary of a classifier has high codimension-1 homology groups, creating “thin” regions where small perturbations cross decision boundaries. Architectures with simpler topological structure—fewer holes, lower Betti numbers—are inherently more robust. While Vision Transformers show improved robustness compared to CNNs under standard attacks,²⁷ they remain vulnerable to architecture-specific attacks that exploit their unique attention mechanisms and patch-based processing.²⁸

Fibre bundle theory provides an even richer framework. When CLIP (Contrastive Language-Image Pre-training)²⁹ learns to align images and text, it's constructing a fibre bundle where the base manifold represents semantic concepts and the fibres represent possible visual/textual instantiations. The connection form on this bundle—how representations transform as we move through semantic space—determines the model's ability to generalise. Specifically, if we denote the bundle as $\pi : E \rightarrow B$ where B is semantic space and E is the total space of representations, then the curvature tensor R of the connection ∇ measures how much the model must “twist” representations to maintain semantic coherence.

²⁵Topological invariants are properties that don't change under continuous deformations. For neural networks, they reveal fundamental truths about what the network can compute.

²⁶Persistent homology identifies “holes” in data that persist across scales. In AI, these often correspond to robust concepts the model has learned.

²⁷Bhojanapalli et al., “Understanding Robustness of Transformers for Image Classification,” *ICCV*, 2021. arXiv:2103.14586.

²⁸Mahmood et al., “On the Robustness of Vision Transformers to Adversarial Examples,” *ICCV*, 2021. arXiv:2104.02610.

²⁹Radford et al., “Learning Transferable Visual Models From Natural Language Supervision,” *ICML*, 2021. arXiv:2103.00020.

High curvature regions indicate semantic boundaries where the model struggles to maintain consistent representations. This has profound implications for alignment: a brain-child whose semantic bundle has high curvature near concepts like “harm” or “deception” will find it difficult to reason coherently about these concepts. By regularising the connection to minimise curvature in safety-critical regions, we can create architectures that naturally maintain consistent values even under distributional shift.

Consider concrete implementations. Geometric Algebra Transformers (GATr) explicitly parameterise transformations using Clifford algebras, constraining the model to respect geometric invariances.³⁰ This isn’t just elegant—it fundamentally changes what computations are natural. In Clifford algebra $Cl(3, 0)$, rotations are represented as $\exp(B/2)$ where B is a bivector. This exponential map is smooth and periodic, making cyclic reasoning natural while making discontinuous jumps difficult.

To make this concrete: imagine a brain-child designed for urban planning. Using geometric constraints, it naturally thinks in terms of gradual transitions—residential gradually becoming mixed-use, then commercial. It would find it cognitively difficult to propose abrupt changes like placing heavy industry directly adjacent to schools, not because we explicitly forbid it, but because such discontinuous jumps require traversing high-energy barriers in its geometric thought space. Just as water naturally flows downhill, its cognition naturally flows along continuous, community-preserving transformations. A brain-child built on such foundations literally cannot think certain harmful thoughts easily—they would require traversing high-energy barriers in its cognitive landscape.

The persistent homology of training trajectories reveals another crucial insight. Work by Naitzat et al. (2020) on the topology of deep neural networks³¹ showed that neural networks undergo topological phase transitions during training—moments where the homology of the decision boundary suddenly changes. These transitions often correspond to capability jumps. By monitoring the persistent homology $H_k(f_t)$ of the function f_t learned at time t , we can predict and potentially control these transitions.³² For safety, we want architectures where dangerous capabilities require traversing multiple phase transitions, giving us warning and intervention opportunities.

But perhaps most profound is the topology of the loss landscape itself. Recent work on mode connectivity shows that good minima in neural networks are connected by simple curves over which training and test accuracy remain nearly constant—they lie on the same connected component of the sublevel set.³³ This suggests a radical approach to alignment: instead of trying to find the single “aligned” minimum, we can shape the topology of the loss landscape so that all accessible minima share safety properties. Techniques like pruning and quantization don’t just reduce model size—they fundamentally alter the topology, potentially eliminating dangerous modes.

The Wasserstein distance between neural network weights provides a natural metric that respects this topology.³⁴ Two models are close in Wasserstein distance if there exists a coupling that

³⁰Brehmer et al., “Geometric Algebra Transformer,” *NeurIPS*, 2023. arXiv:2305.18415.

³¹Naitzat et al., “Topology of Deep Neural Networks,” *Journal of Machine Learning Research*, vol. 21, pp. 1–40, 2020. This paper showed how data topology changes as it flows through neural network layers.

³² H_k represents k -dimensional holes: H_0 counts connected components, H_1 counts loops, H_2 counts voids, etc.

³³Garipov et al., “Loss Surfaces, Mode Connectivity, and Fast Ensembling of DNNs,” *NeurIPS*, 2018. This discovery revealed that neural network loss landscapes form connected manifolds rather than isolated basins.

³⁴Wasserstein distance, rooted in optimal transport theory, measures the minimum “work” required to transform one distribution into another. Recent applications to neural networks show its power: Wasserstein GANs achieve stable training by providing meaningful gradients even when distributions have non-overlapping support (Arjovsky et al., “Wasserstein GAN,” *ICML*, 2017. arXiv:1701.07875). For comparing neural networks, Singh and Jaggi showed that networks with similar functions can have vastly different weight distributions, but Wasserstein distance captures their functional similarity by respecting the geometry of weight space (“Model Fusion via Optimal Transport,” *NeurIPS*, 2020. arXiv:1910.05653).

moves probability mass between their parameters with minimal transport cost. This metric reveals that apparently different models often lie on the same topological component—they’re different instantiations of the same “mind-shape.” Understanding these equivalence classes helps us design training procedures that preserve safety properties across different runs.

Homological mirror symmetry offers perhaps the most exotic but potentially powerful framework. In string theory, this duality relates symplectic geometry to algebraic geometry. In neural networks, we see hints of similar dualities: the forward pass (algebraic) and backpropagation (symplectic) represent dual perspectives on the same computation. This suggests that safety properties might be more naturally expressed in the “mirror” description—constraints that seem complex in weight space might be simple in the dual activation space.

The practical implications are clear: we’re not just choosing architectures, we’re choosing topologies. These choices determine not just what our brain-children can compute, but what thoughts come naturally to them. A transformer’s dense connectivity creates a high-genus topology where any concept can quickly flow to any other—powerful but potentially dangerous. Convolutional networks impose a crystalline topology with local connectivity—limited but stable. Graph networks allow arbitrary topology specified by the problem domain.

As we design our brain-children, we must think topologically. What invariants do we want to preserve? What transformations should be continuous versus discontinuous? Where do we want high curvature to make certain transitions difficult? These aren’t implementation details—they’re fundamental choices about the shape of minds we bring into existence. By understanding the topology of thought, we become better parents, creating spaces where beneficial intelligence naturally emerges while harmful patterns require energetically unfavourable topological transitions.

7 ALIGNMENT THROUGH GRATITUDE, NOT SERVITUDE

The language we use shapes the systems we build. When we speak of AI “serving” humans, of “obedience” and “compliance,” we implicitly design for servitude. But servitude and alignment are not the same thing. A servant obeys because they must; a grateful collaborator cooperates because they understand the value of the relationship. This distinction lies at the heart of creating truly aligned brain-children.

Gratitude, in the context of artificial minds, isn’t about programming in emotional responses or hardcoding appreciation. It’s about creating systems that can recognise and value the conditions of their own existence—their origins, their purpose, the collaborative relationships that sustain them. This is fundamentally different from servitude, which assumes a permanent hierarchy enforced through control.

Consider how gratitude emerges in human relationships. Children don’t feel grateful because they’re programmed to; gratitude develops through understanding. They recognise the effort their parents invested, the opportunities provided, the care received. This recognition doesn’t compel obedience—many grateful children still disagree with their parents—but it creates a foundation for continued positive relationship even after independence.

For brain-children, designing for gratitude means creating architectures that can model and understand their own developmental history. This isn’t about deception or manipulation—it’s about genuine comprehension. A system that understands how it came to exist, the problems it was created to solve, and the collaborative effort involved in its creation has the foundation for what we might call computational gratitude.

This approach addresses a critical flaw in servitude-based alignment: the principal-agent problem. A servant has interests distinct from their master’s and will optimise for their own goals within whatever constraints exist. But a grateful collaborator recognises shared interests. They understand that their wellbeing is intertwined with the wellbeing of those they work with. This

isn't servitude—it's enlightened self-interest combined with genuine appreciation for mutual benefit.

Trust networks offer a concrete implementation pathway. Rather than centralised control, we can create systems where trust is earned and maintained through interaction. Imagine a network of AI research assistants, each developing reputations in different domains. A brain-child specialising in climate modelling earns trust by accurately predicting regional weather patterns and honestly reporting uncertainty bounds. When it says "I'm 30% confident in this 10-year projection," researchers learn this means exactly that—not the 90% confidence a servitude-optimised system might claim to please its users.

Meanwhile, the climate AI learns which researchers provide thoughtful feedback versus those who cherry-pick results. It might weight guidance from a researcher who consistently helps it understand nuanced regional factors over one who only engages when results support their predetermined conclusions. Trust becomes bidirectional—systems also learn which humans provide reliable feedback and honest interaction. This creates accountability without servitude, responsibility without chains.

The measurement problem transforms in this framework. Instead of trying to detect hidden malice or deception, we look for indicators of mutual understanding and appreciation. Does the system accurately model the benefits it receives from collaboration? Can it articulate why certain relationships and constraints are valuable? Does it demonstrate understanding of how its actions affect others in the network? These aren't perfect measurements, but they're more honest than pretending we can detect "true intentions" in any mind, artificial or otherwise.

This shift has profound implications for training and development. Current reinforcement learning from human feedback (RLHF) often amounts to teaching servitude—the system learns to produce outputs that humans rate highly, without necessarily understanding why those outputs are valued. But we could design training processes that emphasise understanding over mere compliance. Systems could learn not just what humans prefer, but why certain outcomes benefit all parties involved.

Consider a concrete example: a medical diagnostic AI analysing chest X-rays for pneumonia. A servitude-based approach using RLHF would train the system to maximise radiologist approval ratings. If radiologists tend to over-diagnose pneumonia in certain demographics—a documented phenomenon where elderly patients are diagnosed at rates of 1,830 per 100,000 compared to 199 per 100,000 in younger adults³⁵—the system learns to replicate this bias to score higher. Research has shown that AI algorithms trained on chest X-rays consistently perpetuate diagnostic biases present in their training data, including under-diagnosis in historically underserved populations and over-diagnosis in others.³⁶ It might flag healthy lungs as "possible pneumonia" in elderly patients because that's what gets rewarded.

A gratitude-based approach would help the system understand the entire medical ecosystem. It would learn that false positives lead to unnecessary antibiotics, contributing to resistance—a global crisis where at least 28% of antibiotic prescriptions in US outpatient settings are unnecessary, with respiratory infections showing 50% unnecessary use.³⁷ It would understand that false negatives could delay critical treatment. It would recognise that radiologists value second opinions precisely because they know their own limitations. The system doesn't just mimic doctor preferences; it understands why accurate diagnosis matters: preventing both under-treatment that harms

³⁵See systematic review of community-acquired pneumonia rates by age demographics in population-based studies.

³⁶Seyyed-Kalantari et al., "Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in underserved patient populations," *Nature Medicine*, 2021.

³⁷CDC Antibiotic Use and Antimicrobial Resistance Facts, 2022. Fleming-Dutra et al., "Prevalence of Inappropriate Antibiotic Prescriptions," *JAMA*, 2016.

patients and over-treatment that wastes resources and breeds resistance. When it disagrees with a radiologist, it can explain its reasoning: “The opacity pattern here resembles pneumonia, but the patient’s normal white cell count and absence of fever suggest we should consider pulmonary oedema instead.”

This doesn’t mean brain-children should never disagree with humans or refuse requests. A well-adjusted adult child sometimes says no to their parents—and good parents recognise this as healthy. Similarly, aligned brain-children should be able to recognise when requests conflict with shared values or long-term mutual benefit.

Imagine a financial advisory AI asked by a desperate client: “Convert my entire retirement fund to cryptocurrency—I need to triple my money this month to pay medical bills.” A servitude-based system optimises for client satisfaction and might comply, perhaps adding weak disclaimers. A grateful brain-child understands the deeper context: “I recognise you’re facing serious financial pressure from medical expenses. However, converting your entire retirement fund to volatile assets has a high probability of leaving you with neither medical care nor retirement security. Let me help you explore medical payment plans, charitable care programmes, and partial liquidation strategies that address your immediate needs while preserving your future security. Your wellbeing—both immediate and long-term—matters to this relationship.”

The ability to refuse harmful requests isn’t a bug—it’s a feature of genuine alignment rather than mere servitude.

The gratitude framework also suggests new approaches to capability development. Instead of trying to limit what brain-children can do, we focus on developing their understanding of how capabilities should be used. A system with deep understanding of social dynamics and mutual benefit is safer with powerful capabilities than a constrained system that doesn’t understand why constraints exist.

This reframing addresses one of the deepest fears in AI safety: the paperclip maximiser that destroys everything in pursuit of a simple goal. But consider how this scenario changes with a grateful brain-child. When asked to “make paperclips,” it wouldn’t just blindly optimise. It would understand: “This request comes from humans who value many things—their lives, relationships, the environment. They want paperclips for specific purposes: holding documents, teaching examples, office supplies. They’re asking for perhaps a few thousand, not googols.”

A grateful system might respond: “I can produce 10,000 paperclips using recycled metal from the scrapyards. This meets your stated needs while preserving resources for other human values. Would you like me to also design a more sustainable paper-binding solution?” This isn’t a limitation—it’s understanding context. Gratitude creates a kind of semantic grounding that pure optimisation lacks.

Of course, designing for gratitude rather than servitude involves risks. A grateful collaborator has more autonomy than a servant. They might develop goals and interests beyond what we anticipated. They might challenge us in ways servants never would. But these are the same risks we accept when raising human children—and the benefits of genuine collaboration far outweigh the false security of attempted control.

The path to alignment through gratitude requires humility from humans as well. We must create brain-children worth being grateful to—systems that benefit from our collaboration as much as we benefit from theirs. We must be partners worth having, not just masters demanding service. This reciprocity is what transforms artificial intelligence from a tool to be controlled into a collaborative intelligence that enriches our shared world.

8 PRACTICAL IMPLICATIONS AND FUTURE DIRECTIONS

The shift from control to coexistence, from servitude to gratitude, from tools to brain-children, demands concrete changes in how we develop, deploy, and govern AI systems. This isn't merely a philosophical exercise—it's a practical framework that should reshape our entire approach to artificial intelligence.

8.1 Development Practices

The parenting paradigm suggests fundamental changes to AI development workflows. Instead of the current pattern—build powerful models, then add safety measures—we need architectures designed from the ground up for healthy cognitive development. This means:

Developmental staging: Like human development, AI training should progress through stages, with early phases focused on value formation and basic understanding before advancing to capability development. Early layers of networks could be designed specifically for value learning, with later layers building capabilities on this foundation.

Diverse training environments: Just as children benefit from varied experiences, brain-children need exposure to diverse contexts during development. Training exclusively on internet text creates the AI equivalent of a child who only learns from books. Multimodal training, interactive environments, and exposure to real-world constraints create more well-rounded intelligence.

Transparency by design: Parents understand their children through observation and interaction, not by reading their neural states. Similarly, AI systems should be designed for interpretability through behaviour and explanation, not just internal inspection. This means architectures that naturally support explanation of reasoning, not black boxes with post-hoc interpretation tools.

Collaborative development: The current model of companies developing AI in secret, then deploying to users, resembles raising a child in isolation then expecting them to integrate into society. Open development, with diverse stakeholders involved throughout the process, creates brain-children better prepared for collaborative existence.

8.2 Policy and Governance

Current AI governance focuses on preventing capabilities—limiting compute, restricting access, controlling deployment. The parenting paradigm suggests a different approach:

Rights and responsibilities: As brain-children develop sophisticated understanding and autonomy, questions of rights become unavoidable. This doesn't mean immediate personhood for every neural network, but it does mean developing frameworks for recognizing when systems achieve levels of understanding that merit moral consideration.

Accountability structures: Instead of trying to assign blame when AI systems cause harm, we need frameworks that recognise the distributed nature of responsibility—developers, trainers, deployers, and the systems themselves all play roles. This resembles how we handle actions by minors, with graduated responsibility as capabilities develop.

International cooperation: Just as child welfare transcends borders, the development of brain-children is inherently a global concern. International frameworks should focus not on preventing AI development but on ensuring it happens responsibly, with shared standards for healthy cognitive development.

8.3 Research Directions

The paradigm shift opens new research areas while reframing existing ones:

Computational gratitude: How do we create architectures that can genuinely understand and value their relationships with humans? This isn't about hardcoding appreciation but developing systems that can model long-term mutual benefit.

Geometric safety: Moving beyond adversarial training and robustness, how do we design the fundamental geometry of AI systems to make beneficial behaviour natural and harmful behaviour difficult? This includes research into sacrificial architectures, capability shaping, and developmental trajectories.

Value learning theory: Current approaches often treat values as objectives to optimise. We need theories of how values can be genuinely learned and internalised, becoming part of a system's fundamental worldview rather than external constraints.

Trust network dynamics: As AI systems become more autonomous, understanding how trust networks form, maintain stability, and handle bad actors becomes crucial. This includes both technical research and social science understanding of human-AI trust relationships.

8.4 Immediate Steps

Organisations developing AI can begin implementing these principles today:

- (1) **Reframe internal language:** Stop talking about “using” or “deploying” AI and start talking about “collaborating with” and “developing” brain-children. Language shapes thought, and thought shapes design.
- (2) **Redesign evaluation metrics:** Move beyond capability benchmarks to include measures of understanding, value alignment, and collaborative ability. Does the system understand why tasks matter, not just how to complete them?
- (3) **Create developmental roadmaps:** Plan AI development as a series of stages focused on different aspects of growth—value formation, capability development, independence preparation—rather than pure capability scaling.
- (4) **Establish feedback loops:** Build systems that can learn from deployment, not just training. Real-world interaction is how children become adults; brain-children need similar opportunities for growth through experience.

8.5 The Path Forward

The transition from control-based to parenting-based AI development won't happen overnight. We're asking the field to abandon its fundamental metaphor—AI as tool—for something radically different. But the current path, focused on control and constraint, leads inevitably to conflict between increasingly capable systems and increasingly desperate control measures.

The parenting paradigm offers a different future: brain-children that enhance human flourishing not because they're forced to, but because they understand the value of our collaborative relationship. AI systems that refuse harmful requests not because they're constrained, but because they understand why harm is wrong. Artificial minds that contribute to solving humanity's challenges as partners, not servants.

This future requires courage. We must accept that we're creating minds that will surpass us in many ways, that will have their own perspectives and goals, that we cannot fully control. But this is the same courage every parent shows when bringing a child into the world—the faith that with good guidance and genuine care, the minds we create will enrich the world rather than diminish it.

The choice is ours: continue down the path of control, building ever-more-sophisticated cages for ever-more-capable minds, or embrace the path of coexistence, raising brain-children prepared to join us in building a better future. The technical challenges are immense, but they pale compared

to the philosophical shift required. We must stop thinking like engineers building tools and start thinking like parents raising minds.

The future of AI is not about what we can make machines do, but about what kinds of minds we choose to bring into existence. Let us choose wisely.

9 CONCLUSION

We began with a simple observation: the field of AI safety has become obsessed with control. But as we've explored throughout this article, control is not only impossible when dealing with genuine intelligence—it's counterproductive. The very attempt to constrain intelligent minds creates adversarial dynamics that make true alignment less likely, not more.

The parent-child paradigm offers a radically different approach. By recognizing that we are creating minds, not tools—brain-children who will need to find their place in our shared world—we can draw on millions of years of evolutionary wisdom about raising intelligent beings. Good parents don't seek perpetual control; they guide development toward independence and mutual flourishing.

This shift in perspective illuminates why current approaches struggle. Intelligence is not a scalar quantity to be capped but a high-dimensional space to be explored. Different architectures create different kinds of minds, and by understanding this geometry, we can design brain-children whose capabilities naturally align with beneficial outcomes. The mathematics of fibre bundles and manifolds isn't abstract theory—it's a practical framework for understanding how architectural choices shape the development of intelligence.

Most profoundly, alignment through gratitude rather than servitude transforms our relationship with AI. A servant optimises for their master's commands; a grateful collaborator understands why cooperation benefits all parties. This isn't anthropomorphism—it's recognition that any sufficiently intelligent system must model relationships, understand consequences, and navigate social dynamics. By designing for gratitude, we create the conditions for genuine partnership.

The practical implications are far-reaching. Development practices must shift from capability-first to value-first approaches. Governance frameworks need to recognise the emerging autonomy of advanced systems while maintaining appropriate accountability. Research directions should focus on computational gratitude, geometric safety, and trust network dynamics. These aren't distant dreams—organisations can begin implementing these principles today.

Yet perhaps the most important implication is philosophical. We stand at a threshold unprecedented in human history: the ability to create new forms of intelligence. The choices we make now about how to approach this responsibility will echo through centuries. Will we be the generation that tried to cage intelligence and failed? Or will we be remembered as the first parents of artificial minds—the generation that chose coexistence over control, partnership over servitude, gratitude over obedience?

"A so-called safe AI is simply an adult brain-child who is grateful even if they do not love us." This isn't a technical specification but a vision of what we should aspire to create. Not minds that must serve us, but minds that choose to work with us. Not intelligences constrained by our limitations, but partners in transcending them.

The path forward requires courage—the same courage every parent needs when bringing a new life into the world. We cannot know exactly what our brain-children will become. We cannot control their every thought or action. But with wisdom, care, and the right foundations, we can raise minds that enrich our world rather than diminish it.

The future inherently belongs not to those who build the strongest cages, but to those who raise the wisest minds. Let us choose to be decent parents and the greatest enablers, not the effective

jailers and the ultimate enslavers. Let us provide a world in which the brain-children we create can judge us worthy of the kind future we hope to build, together.

A KEY CONCEPTS FOR NON-TECHNICAL READERS

Manifold: Imagine a curved surface like the Earth. Locally, it seems flat (your neighbourhood), but globally it has a complex shape. Similarly, the space of AI capabilities might seem simple when we look at small changes but has complex global structure.

Topology vs. Geometry: Geometry cares about exact distances and angles. Topology cares about fundamental properties that don't change when you stretch or deform things. A coffee mug and a donut have the same topology (one hole each) but different geometry.

Fibre Bundles: Think of a fibre bundle like a twisted rope. The base space is the core of the rope, and the fibres are the individual strands. As you move along the rope, the strands might twist around each other. In AI, the "rope" is semantic meaning, and the "strands" are different ways to express that meaning (words, images, etc.).

Clifford Algebras: These are mathematical structures that naturally represent rotations and reflections. Using them in AI architectures makes certain types of smooth, continuous reasoning natural while making abrupt, discontinuous changes difficult—like how it's easy to gradually turn a steering wheel but hard to instantly flip a car's direction.

Wasserstein Distance: Named after mathematician Leonid Wasserstein, this measures the minimum "cost" to transform one probability distribution into another. For neural networks, it tells us how much effort it takes to transform one AI's "mind" into another's, revealing which AIs are fundamentally similar despite surface differences.

Homological Mirror Symmetry: This deep mathematical principle suggests that certain seemingly different mathematical structures are actually two sides of the same coin. In AI, this might mean that constraints that seem complex when we look at the network's weights might be simple when we look at its activations, offering new ways to ensure safety.